

KUSHAGRA AGRAWAL

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EDUCATION

Bachelor of Computer Applications (BCA)

Amity University

2024 - 2027

Noida, Uttar Pradesh

- Cumulative GPA: 9.08/10.0
- Relevant Coursework: Algorithms and Data Structure, Machine Learning, Deep Learning, Probability and Applied Statistics, Applied Mathematics

WORK EXPERIENCE

AI & Data Science Intern, Unihox

July 2025 - Dec 2025

Pune, Maharashtra

- Built a **multi-agent AI system** coordinating research, data fusion, and decision-making agents to match patients with medical specialists based on symptoms, medical history, and availability, **achieving 95%** user satisfaction.
- Engineered backend infrastructure supporting **500+ patient profiles**, with **PostgreSQL** for structured data, **MongoDB** for medical records, and **Pinecone** for semantic search across **5,000+ medical documents**.
- Optimized query performance through Redis caching and connection pooling, reducing average latency to **under 100ms** for real-time concurrent requests.
- Created a comprehensive suite of unit and integration tests covering **85%** of the codebase, improving code quality and catching bugs early in the development lifecycle.

PROJECTS

AgentCost [Link](#)

- Architected an open-source observability platform supporting **1,900+ models**, designed to scale to **50M+ tokens** with **<50ms latency** on complex analytical queries.
- Engineered a high-performance Python SDK with **zero-code LangChain integration** using runtime monkey-patching, benchmarking **<1ms overhead** while capturing full request/response traces.
- Developed an async FastAPI backend with PostgreSQL and pre-aggregated analytics, designed and load-tested to handle **100K+ daily events** without performance degradation.
- Built a **self-optimizing ML engine** using **statistical anomaly detection (Z-scores)** and **feedback-driven learning**, automatically ranking model alternatives based on historical success rates and cost-effectiveness.

NexusAgent.ai [Link](#)

- Developed an **AI interview agent platform** using **Google AI SDK** and **Vapi AI**, delivering real-time voice-interactive mock interviews with instant performance analysis.
- Integrated **Vapi Web SDK** for natural voice interactions, that simulate realistic interview scenarios with live speech recognition, and smart AI-generated follow-up questions based on responses.
- Implemented **Firestore backend** with **Auth** and **Admin**, supporting **50+ interview sessions** with real-time data management and user authentication.
- Automated engagement system using **GitHub Actions cron jobs** for email reminders (**24h pre-interview**), weekly PDF analytics reports, and streak tracking notifications.

CERTIFICATIONS & SKILLS

- Certifications:** Deep Learning Specialization, Generative AI for Everyone, Python for Data Science, AI & Development
- Skills:** Python, SQL, JavaScript, TensorFlow, PyTorch, Transformers, CNN, NLP, NLU, TF-IDF
- Technologies/Frameworks:** AWS, Snowflake, FastAPI, Docker, DigitalOcean, RAG, MLflow, Git, GitHub, Hugging Face, Plotly, BERT, ReAct